

FoldShield⁺⁺ Structural Analysis Report

Generated 2026-06-04 15:22 · Mode: ca · RexCrux Research Laboratory

Input Structures

Reference PDB	1A3N.pdb
Query PDB	1A00.pdb
Mode	ca
Ref chain	A
Query chain	A

Classical Coordinate Metrics

These metrics saturate near 1.0 even when function diverges. Use FoldShield++ signals below for structural perturbation insight.

Metric	Value	Note
TM-score	1.000	>=0.50 same fold · >=0.90 nearly identical
RMSD (A)	0.300	Lower is better · >3.0 A = significant deviation

FoldShield++ Signal Scores — Phase 1 + Phase 2

All active signals with scores, fusion weights, and interpretation. Phase 2 signals are highlighted in blue.

Braid similarity	0.673	0.20	moderate braid similarity
Motif similarity (UL-DSL)	0.950	0.37	nearly identical motif architecture
SSEF entropy (Phase 2) [P2]	0.974	0.20	symbolic grammar nearly identical — buried-core mutation or identical
Persistent homology (Phase 3) [P2]	1.000	0.23	persistent homology features nearly identical

Fusion Layer v2 — Combined Score

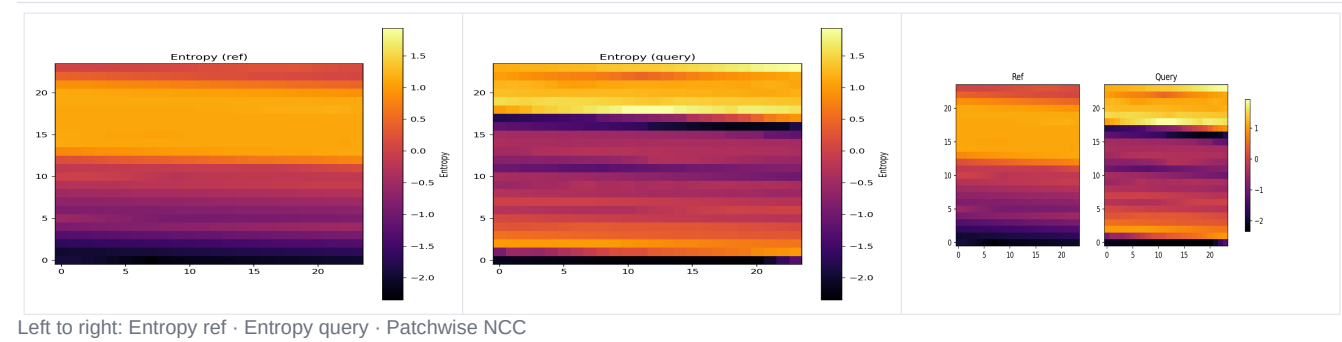
0.9097	phase2	nearly identical protein structures

Phase 2 — Perturbation Classification

SSEF score	Local topology	Gap score	Perturbation class
0.974	0.630	0.344	Conformational transition

Gap = SSEF minus local topology. <0.15 buried-core mutation | 0.15-0.36 fold-switch | >0.36 conformational transition

Entropy Maps & Signal Profiles



Signal Score Bars

Visual 0-1 scale for all active signals. Green ≥ 0.66 / Amber 0.33-0.66 / Red < 0.33 .



Symbolic Score Detail

Symbolic Similarity Score	0.673	overall strong similarity
Behavioural Correlation Similarity	0.487	moderate entropy similarity
Structural Topology Similarity	0.993	identical set of generators
Structural Topology Ordering Similarity	0.660	strong pattern similarity

Signal Analysis — Figures

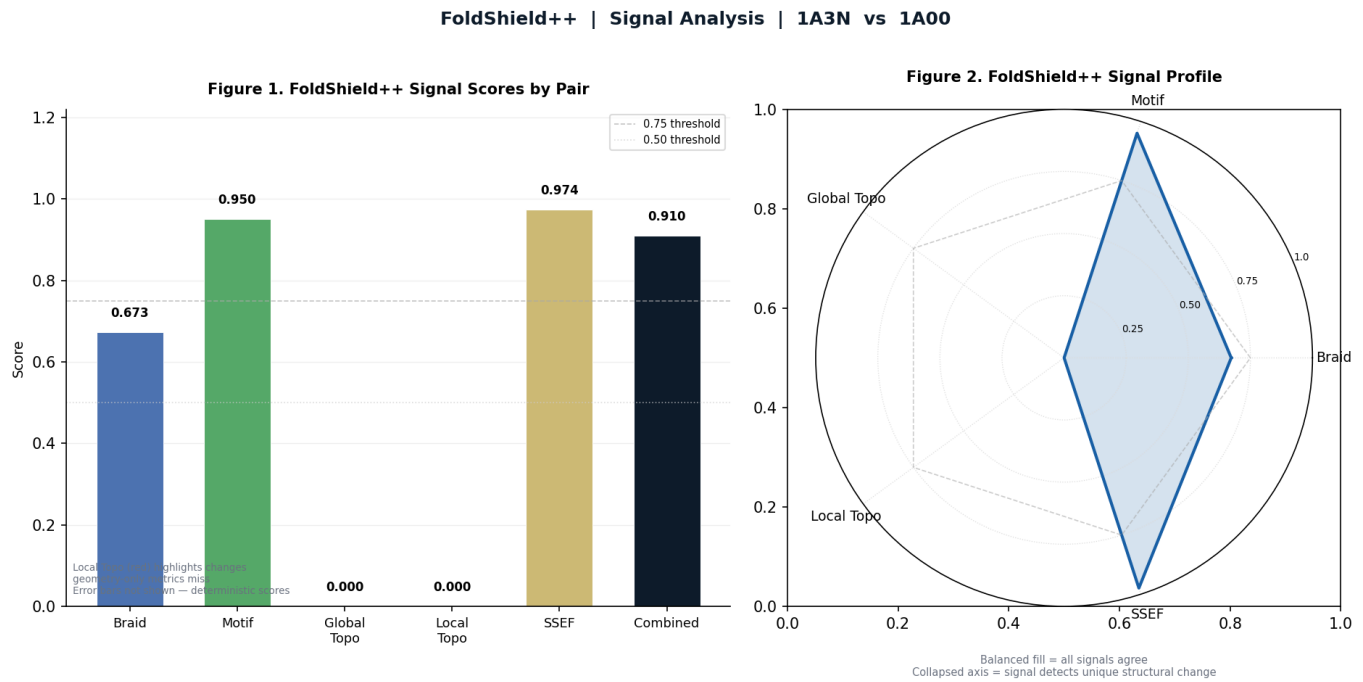


Figure 1: Bar chart comparing all five signals and the Combined score. Local Topo (red) highlights structural changes geometry-only metrics miss. Dashed = 0.75 · Dotted = 0.50. Error bars not shown — scores are deterministic. Figure 2: Radar chart showing the five-signal profile. Balanced fill = all signals agree. Collapsed axis = signal detects unique structural change.